

Yafira Martinez

Design Engineer / Creative Technologist

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Interactive Systems, Interfaces & Experimental Technologies

SKILLS

Front-End Development: React, Next.js, TypeScript, JavaScript, HTML/CSS, Node.js, REST APIs, Web Serial, p5.js, ml5.js, Creative Coding, Supabase, Vercel, Git

Design: Interaction Design, UX/UI, Responsive Web Design, Design Systems, Prototyping, Wireframing, Digital Accessibility, Information Architecture, User Research, Figma, Adobe Suite

Hardware: Physical Computing, Microcontrollers (Arduino, Raspberry Pi, ESP32, RP2040, Adafruit), CircuitPython, Python, Fritzing, E-Textiles, Soft Circuits, Wearables, Soldering, Fabrication (laser cutting, 3D printing, knitting machine)

EXPERIENCE

Design Engineer & Creative Technologist — [Electrocute Lab](#) Mar 2021 – Present
Creative independent practice

- Design and ship independent web applications and tools end to end — electrocute-ui, a React component library with an 18-color design token system published to npm, and tinytinker.tools, an open-source maker toolkit of 22+ browser tools — alongside maker tutorials, open workshop materials, and project documentation on a technical blog (Astro + TinaCMS).
- Work at the intersection of textile craft and computation — soft circuits, e-textiles, physical computing, and generative text — with projects exhibited at venues including the Open Hardware Summit in Berlin and taught in public workshops.

Lead Camp Counselor — [ITP Camp, New York University](#) Summer 2026
Previously: Camp Counselor, Summer 2025

- Led a 150+ person creative technology community through workshops, mentorship, and hands-on support across hardware, software, and fabrication.
- Built spot-mapper, a real-time collaborative floor-plan mapping tool (Leaflet, Vercel serverless functions, Redis, token-based auth) used live during the Show All Things Show, with admin/viewer roles, mobile support, and full handoff documentation.

Hardware Research Developer — [NYU Ability Lab](#) Oct 2025 – May 2026

- Researched accessible fashion technologies for the Making Fashion Accessible project, using knitting machines and AYAB (open-source knitting machine software) to support inclusive, open-source garment production and adaptable textile workflows.

Graduate Teaching & Research Assistant — [NYU ITP/IMA](#) Sep 2025 – May 2026

- Graduate Teaching Assistant for Creative Computing (Prof. Daniel Shiffman) and Intro to Wearables (Instructor Daniel Ryan Johnston), mentoring students in creative coding and wearable design; Research Assistant for Computational Text, developing syllabus material spanning procedural text, data ethics, and machine learning.
- Designing a Teachable Machines workshop — ml5.js + p5.js image-classification games — to be co-led for ~60 middle school students with NYU's College & Career Lab (upcoming, Jul 2026).

Machine Learning Researcher & Web Contributor — [ml5.js](#) Jun 2025 – May 2026

Mentors: Daniel Shiffman & J.H. Moon

- Researched and prototyped ml5.transcribe() and ml5.embeddings(), bringing browser-based transformer models (Whisper, Moonshine) to a library designed for students, artists, and beginner coders.
- Refined interaction design and inference performance, and contributed documentation and examples to support contributor and learner usability.

Web Developer & Designer — [Useless Machines Archive, NYU](#) Mar 2025 – Aug 2025

Professor: Blair Simmons

- Designed and built a front-end web archive using the Are.na API, focusing on client-side data fetching, content organization, and responsive interface design.

Lead Web Developer — [Equitable Syllabus Project, NYU ITP/IMA](#) Nov 2024 – May 2026

- Redesigned and rebuilt the project website with accessible, user-centered interfaces — from Figma designs to a production Next.js deployment — while migrating project data and maintaining centralized contribution tracking.

Software Engineer & Creative Technologist Intern — [Floreo Labs](#) Jan 2020 – May 2020

- Developed responsive front-end applications in React and designed UI/UX prototypes in Adobe XD; contributed to mobile projects while managing agile workflows and producing technical documentation.

EDUCATION & CERTIFICATIONS

Master of Professional Studies (MPS), Interactive Telecommunications Program May 2026
[New York University, Tisch School of the Arts](#)

Bachelor of Science (BS), Computer Science Dec 2020
[City University of New York, Lehman College](#)

Certifications: Artificial Intelligence, Samsung Innovation Campus (2022–2023) · Human Computer Interaction, Interaction Design Foundation (2022) · Google UX Design, Coursera (2021)

SELECTED WORK & EXPLORATIONS

- Soft Circuit Boards (SCBs) Workshop** — [ITP Camp, NYU](#) Jun 2026
Designed and taught a hands-on workshop on handmade circuits — copper tape traces, craft foam boards, and LEDs, no soldering required — extending work from the Soft Circuit Library.
- PetalByte** — [Bring Your Own Data \(BYOD\), NY Tech Week by a16z, NYC](#) Jun 2026
Portable generative text device treating the USB drive as both archive and interface; Markov chain corpora live on a flower-shaped laser-cut acrylic enclosure, browsable through a custom pygame interface on a Raspberry Pi touchscreen.
- The Soft Computer Mini** — [Open Hardware Summit, Berlin](#) May 2026
Travel-ready extension of The Soft Computer, exploring embodied input, generative text, and textile as substrate, compressed into a tarot-sized device.
- The Soft Computer** — [ITP Thesis, NYU](#) May 2026
Textile-based computing interface combining machine-knit wool, capacitive fabric inputs, haptics, and a flexible e-ink display; Raspberry Pi 5 drives Markov chain text generation across four corpus-mapped channels.
- Open Hardware Summit Fellow** — [Open Source Hardware Association, Berlin](#) Feb 2026 – Jun 2026
Selected fellow exhibiting The Soft Computer and other experimental open-source hardware projects at the Open Hardware Summit, Berlin.
- Soft Circuit Library** — [Electronics Faire \(Theme: Repair\), Temple University Libraries](#) Apr 2026
Collection of handmade textile circuit boards organized as a living archive, each designed to be handled, studied, and repaired by touch.
- Guardians of the Groceries** — [Best of Software, PlayfulNYC Hackathon](#) Apr 2026
Awarded Best of Software. Built the hardware and software layer of a handheld ESP32 device that captures photos, uploads to Supabase via WiFi, and pairs with a live web registry documenting NYC bodega cats.
- Pocket Oracle of Time** — [Horological Society of New York](#) Feb – Apr 2026
Poetic timekeeping object exploring ritual, slowness, and material interaction with time through embedded electronics and tactile design.
- Enoki** — [MIT Media Lab AI × Hardware Hackathon](#) Mar 2026
Focus companion for communal accountability: a React Native app using Mentra smart glasses and Claude Vision AI paired with a Raspberry Pi desktop device that visualizes focus as a mushroom that grows when on task and wilts when distracted.
- Moon Pocket** — [Pedestal Playtime: Art Toy & Tech Toy Open Mic, NYC Resistor](#) Jan 2026
Connected device (Arduino Nano 33 IoT) that fetches real-time moon and tide data from public web APIs and renders it on a round TFT display through touch-based interaction.
- Tendertronics** — [NYU ITP NIME Show, Littlefield](#) Dec 2025
Analog instrument performance exploring tactile, minimal interfaces and embodied sound interaction through handcrafted circuitry.
- Lightning Talk Speaker** — [Exploring Creativity in the Age of AI, NYU Global Show & Tell](#) Oct 2025
Presented "Open Tools, Soft Interfaces" on ml5.js contributions and framing creativity in AI through openness, design as dialogue, and soft, human-centered interfaces.
- Static Chic** — [Volt Couture: Wearables + E-textiles Open Mic, NYC Resistor](#) May 2025
3D-printed inductive handbag with an embedded coil that wirelessly powers and lights up LEDs, presented at an open mic curated by Kari Love and Liza Stark.
- MIT Reality Hack** — [Massachusetts Institute of Technology](#) Jan 2025
Implemented sound and haptic feedback for HangarPro VR, a virtual reality training simulation using Cognitive3D for pre-flight safety.